

The quadratic formula

ANSWER KEY • FOR TEACHERS



Interactive version

CHECK

Check yourself

Level 1 • threshold

1.

How many real roots does $3x^2 - 5x + 4 = 0$ have?

- ☐ A one
 ☒ B none — $D = (-5)^2 - 4 \cdot 3 \cdot 4 = 25 - 48 = -23 < 0$.
 ☐ C can't tell without solving
 ☐ D two

Level 1 • threshold

2.

Solve $x^2 - 5x - 14 = 0$. Give both roots, separated by a semicolon.

WORKED SOLUTION

$$D = (-5)^2 - 4 \cdot 1 \cdot (-14) = 25 + 56 = 81, \text{ so } x_{1,2} = \frac{5 \pm 9}{2}: x_1 = 7, x_2 = -2.$$

Answer: $x_1 = 7, x_2 = -2$

Level 2 • satisfactory

3.

Put the steps for solving $2x^2 + x - 6 = 0$ in the right order.

- 1 Read off $a = 2, b = 1, c = -6$.
- 2 Compute $D = 1^2 - 4 \cdot 2 \cdot (-6) = 49$.
- 3 $D > 0$, so there are two roots.
- 4 $x_{1,2} = \frac{-1 \pm 7}{4}$.
- 5 $x_1 = \frac{3}{2}, x_2 = -2$.

Level 2 • satisfactory

4.

For which value of c does $x^2 + 6x + c = 0$ have exactly one root?**Answer:** 9

5.

Which equations have two different real roots? Choose all that apply.

☐ A $x^2 - 4x + 4 = 0$

☐ B $x^2 + 4 = 0$

☒ C $-x^2 + 3x + 1 = 0$
 $-a$ is negative, but
 $D = 9 + 4 = 13 > 0$.

☒ D $x^2 - 4 = 0$ — $D = 0 - 4 \cdot 1 \cdot (-4) = 16 > 0$: the roots are ± 2 .

EXAM TASK

Two consecutive integers

The sum of the squares of two consecutive positive integers is 181. Find the integers.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

Writes an equation, e.g. $n^2 + (n+1)^2 = 181$. 1 pt

Brings it to the form $n^2 + n - 90 = 0$. 1 pt

Solves it: $n = 9$ or $n = -10$. 1 pt

Rejects $n = -10$ (not positive) and answers: 9 and 10. 1 pt

WORKED SOLUTION

Let the integers be n and $n+1$. Then $n^2 + (n+1)^2 = 181$, so $2n^2 + 2n + 1 = 181$ and $n^2 + n - 90 = 0$.

$$D = 1 + 360 = 361 = 19^2, \text{ so } n = \frac{-1 \pm 19}{2}: n = 9 \text{ or } n = -10.$$

The integers are positive, so $n = 9$. **Answer: 9 and 10.** Check: $81 + 100 = 181$.

TEACHER NOTES

For the teacher

Pupils can say: “I can solve any quadratic equation with the formula, explain where the formula comes from, and tell how many roots an equation has before solving it.”

Common mistakes and what helps

- Sign errors in $-4ac$ when a or c is negative: have pupils write $D = b^2 - 4 \cdot a \cdot c$ with brackets around every coefficient.
- Forgetting the $\pm$, or dividing only $\sqrt{D}$ by $2a$: write the whole numerator over $2a$ first, then simplify.
- Reading a, b, c before moving everything to one side (the garden: $x(x+3) = 30$).
- Reading $D < 0$ as “the roots are negative”.

Key terms (EN · LT): quadratic equation · *kvadratinė lygtis*; discriminant · *diskriminantas*; root · *sprendinys*; coefficient · *koeficientas*; completing the square · *pilnojo kvadrato išskyrimas*.

Timing (2 × 45 min): lesson 1: hook, explore, explain (the proof is required by the programme). Lesson 2: examples, check, exam task.

Programme: 32.1.1.1, “the quadratic formula is proved and applied; the discriminant is linked to the number of roots; word problems are modelled with quadratic equations” (V-594 text). Achievements A1, A3, B2, C1.